

Solution Key: Problem Set 10

CBE 30235

Problem 1: The Inevitable Heat (Decay Heat Physics)

(a) Decay Power Calculation

We use the Way-Wigner approximation:

$$\frac{P(t)}{P_0} = 0.066 [t^{-0.2} - (t + T_0)^{-0.2}]$$

Given: $P_0 = 3000 \text{ MW}_{th}$ and Operating time $T_0 = 18 \text{ months} \approx 4.7 \times 10^7 \text{ s}$.

1. At $t = 1 \text{ second}$:

$$\frac{P}{P_0} = 0.066 [1^{-0.2} - (1 + 4.7 \times 10^7)^{-0.2}] \approx 0.066[1.0 - 0.029] = 0.064$$

$$P(1s) = 3000 \text{ MW} \times 0.064 = \mathbf{192 \text{ MW}}$$

2. At $t = 1 \text{ hour}$ (3600 s):

$$\frac{P}{P_0} = 0.066 [3600^{-0.2} - (3600 + 4.7 \times 10^7)^{-0.2}]$$

$$3600^{-0.2} = 0.1944$$

$$\frac{P}{P_0} = 0.066[0.1944 - 0.029] = 0.0109 \approx 1.1\%$$

$$P(1hr) = 3000 \times 0.0109 = \mathbf{32.7 \text{ MW}}$$

3. At $t = 1 \text{ month}$ ($2.6 \times 10^6 \text{ s}$):

$$(2.6 \times 10^6)^{-0.2} = 0.052$$

$$\frac{P}{P_0} = 0.066[0.052 - 0.029] = 0.0015$$

$$P(1mo) = 3000 \times 0.0015 = \mathbf{4.5 \text{ MW}}$$

(b) Boil-off Rate at 1 Hour

The heat must be removed by the latent heat of vaporization ($h_{fg} \approx 2257 \text{ kJ/kg}$).

$$\dot{m} = \frac{Q}{h_{fg}} = \frac{32,700 \text{ kJ/s}}{2257 \text{ kJ/kg}} \approx \mathbf{14.5 \text{ kg/s}}$$

(c) Fire Truck Capability

A standard fire hose provides $\approx 30 \text{ kg/s}$. Since $30 > 14.5$, a single fire truck provides sufficient volumetric flow to keep the core covered (assuming it can overcome vessel pressure).

(d) Reflood Danger (Critical Thinking)

No, it might not cool it immediately. If the cladding temperature is $> 1200^\circ\text{C}$, the Zirconium-Water reaction ($\text{Zr} + 2\text{H}_2\text{O} \rightarrow \text{ZrO}_2 + 2\text{H}_2$) becomes self-sustaining. This reaction is highly exothermic. Injecting water provides the oxidant (steam) for this reaction, potentially causing a temperature spike ("Reflood Excursion") and massive Hydrogen generation before the cooling effect takes over.

Problem 2: Defense in Depth (PRA)**(a) Fault Tree Structure**

Top Event: Total Loss of AC Power.

$$\text{Top} = \text{LOSP} \cap (\text{EDG}_A \cap \text{EDG}_B)$$

Logic: An AND gate connects the LOSP event and the failure of the backup system. An AND gate connects the two redundant diesels.

(b) Independent Failure Frequency

Assuming independence:

$$f_{\text{indep}} = f_{\text{LOSP}} \times P(A) \times P(B)$$

$$f_{\text{indep}} = 0.1/\text{yr} \times 0.02 \times 0.02 = 4 \times 10^{-5} / \text{year}$$

(Recurrence interval: 1 in 25,000 years).

(c) Common Mode Failure (Tsunami)

Here, the probability of diesel failure given a Tsunami is 1.0.

$$f_{\text{CM}} = f_{\text{Tsunami}} \times P(\text{Both Fail}|\text{Tsunami})$$

$$f_{\text{CM}} = 10^{-3}/\text{yr} \times 1.0 = 10^{-3} / \text{year}$$

Comparison: The Common Mode failure frequency is **25 times higher** than the calculated independent failure frequency. This illustrates why redundant systems fail if they share a common vulnerability.

Problem 3: Reactor Physics (Chernobyl)**(a) Reactivity Insertion**

$$\rho = \alpha_v \times \Delta\text{Void}$$

$$\rho = (+2.0 \times 10^{-4}) \times (40) = 0.008 \Delta\text{k/k}$$

(b) Comparison to β

$$\rho = 0.008$$

$$\beta = 0.0065$$

$$\rho > \beta$$

(c) Prompt Criticality

Since $\rho > \beta$, the reactor is **Prompt Critical**. The power rise is no longer governed by the delayed

neutrons (seconds scale) but by the prompt neutron lifetime ($\approx 10^{-4}$ seconds). The period becomes extremely short (milliseconds), and the power will rise exponentially faster than any mechanical control system (SCRAM) can respond. This leads to a runaway power excursion.

Problem 4: The Hydrogen Problem

(a) Mass of Hydrogen

Mass of Zr oxidized = $0.25 \times 20,000 = 5,000$ kg.

Molar Mass of Zr ≈ 91.2 g/mol.

$$\text{Moles Zr} = \frac{5,000,000 \text{ g}}{91.2 \text{ g/mol}} \approx 54,825 \text{ mols}$$

Reaction: $Zr + 2H_2O \rightarrow ZrO_2 + 2H_2$.

Moles $H_2 = 2 \times \text{Moles Zr} = 109,650$ mols.

Mass $H_2 = 109,650 \text{ mols} \times 2.016 \text{ g/mol} \approx 221,000 \text{ g} = \mathbf{221 \text{ kg}}$

(b) Chemical Heat (Oxidation)

$$Q_{ox} = 5,000 \text{ kg} \times 6.5 \text{ MJ/kg} = 32,500 \text{ MJ} = \mathbf{32.5 \text{ GJ}}$$

Comparison to TNT:

$$\frac{32.5 \text{ GJ}}{4.18 \text{ GJ/ton}} \approx \mathbf{7.8 \text{ tons of TNT}}$$

(c) Chemical Heat (Combustion)

Heat of combustion ≈ 242 kJ/mol.

$$Q_{comb} = 109,650 \text{ mols} \times 242 \text{ kJ/mol} = 26,535,000 \text{ kJ} \approx \mathbf{26.5 \text{ GJ}}$$

(d) Discussion

The energies are of comparable magnitude (32.5 GJ vs 26.5 GJ). This means the "heat source" during a meltdown isn't just the nuclear decay heat; the chemical energy stored in the cladding is massive.

Problem 5: Pressurizing the Containment

(a) Steam Partial Pressure

We assume the entire 250,000 kg inventory flashes to steam (conservative assumption or complete blowdown).

$$n = \frac{250,000,000 \text{ g}}{18 \text{ g/mol}} = 1.39 \times 10^7 \text{ mols}$$

Temp $T = 120^\circ\text{C} = 393 \text{ K}$. Volume $V = 50,000 \text{ m}^3$.

$$P_{steam} = \frac{nRT}{V} = \frac{(1.39 \times 10^7)(8.314)(393)}{50,000}$$

$$P_{steam} \approx 908,000 \text{ Pa} \approx \mathbf{908 \text{ kPa}} \quad (\approx 9 \text{ atm})$$

(b) Total Pressure

We must add the partial pressure of the air, corrected for the temperature rise from 30°C (303 K) to 120°C (393 K).

$$P_{air} = 101 \text{ kPa} \times \frac{393}{303} \approx 131 \text{ kPa}$$

$$P_{total} = P_{steam} + P_{air} = 908 + 131 = \mathbf{1,039 \text{ kPa}} \quad (\approx 10.3 \text{ atm})$$

(c) Survivability

The design pressure is 4 atm ($\approx 400 \text{ kPa}$).

$$10.3 \text{ atm} \gg 4 \text{ atm}$$

The containment would **fail**.

Note: This calculation shows why "Ice Condensers" or "Containment Sprays" are required. They condense the steam back to water to drastically lower P_{steam} and save the building.